Introduction to JavaScript

Assignment 4 report / continuation of the original nine-page website

APEX MOTORS
Interactive features for a considered dealership experience.

Team	Apex Motors
Group	SE-2507
Members	Arsen, Kuttibay, Farabi, Muhammad
Source project	Apex-Motors-Midterm-Project.zip
Technology	HTML, CSS and browser JavaScript; local image assets

Objective and starting point

Extend the existing dealership website with JavaScript form validation, a FAQ accordion, a subscription popup, background color controls, and a live date/time display. Preserve the original page structure, content, images, visual identity and responsive layouts.

The original project contains nine pages. Its contact form had only a native validity check. The updated project uses one shared deferred script, **js/main.js**, linked from every page. All HTML, CSS and JavaScript are readable and contain no code comments.

Where to find the tasks

Task 1	Contact: contact form and Apex Club profile validation
Task 2	Services: four service questions and answers
Tasks 3, 4 and 5	Every footer: Subscribe to Apex, Change background, and local date/time

All forms are front-end demonstrations. They do not create accounts, send requests, save personal information, or store passwords.

Task 1 / Form validation

Required fields, email format, password length and confirmation

1. Keep the original contact form and add a separate Apex Club profile form on Contact.
2. Add labels, required fields, inline error elements and accessible status messages.
3. Intercept submission in JavaScript and validate each field.
4. Focus the first invalid field and recheck corrections as the user types.
5. On success, clear the fields and display an explicit demonstration-only confirmation.

Name and email	Required; names containing only spaces are rejected. Email must contain a valid local part, @, and a domain.
Password	At least 8 characters; whitespace-only values are rejected.
Confirmation	Required and must match the original password exactly.

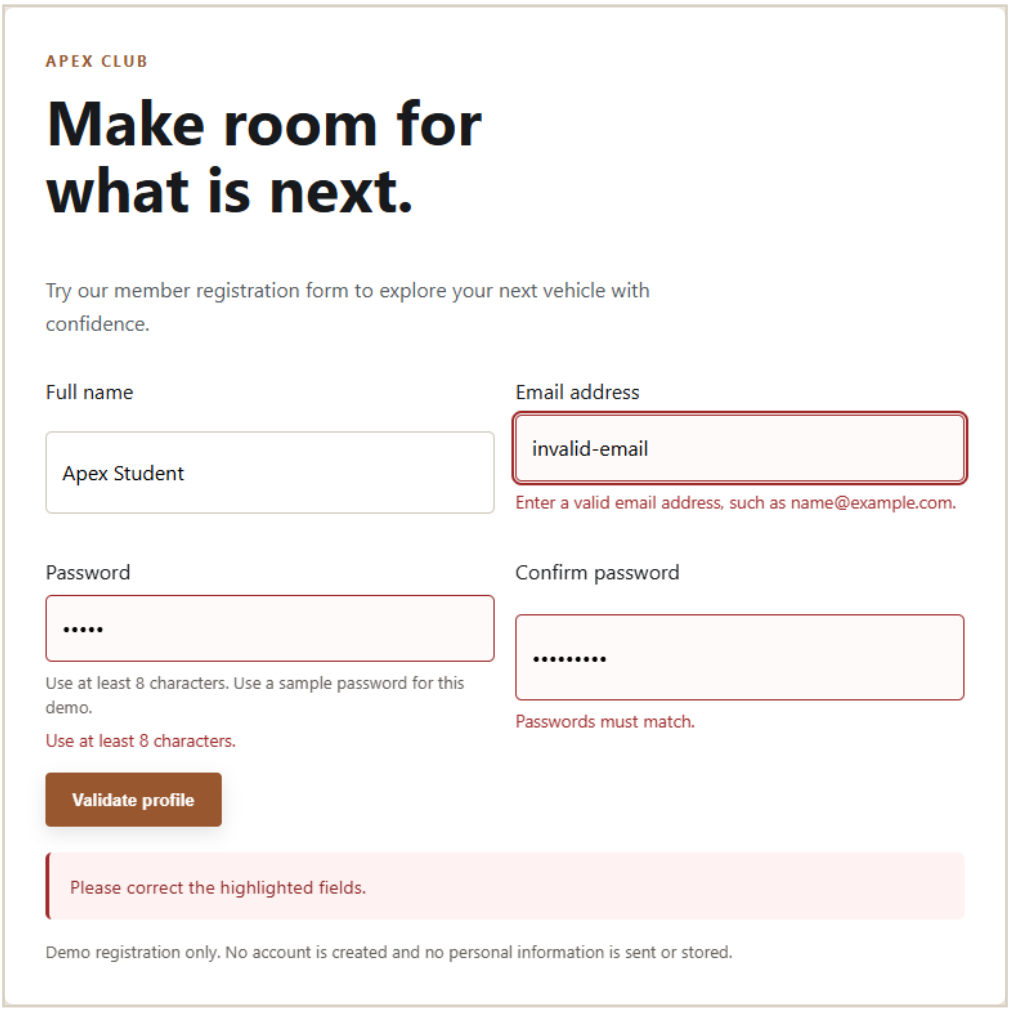


Figure 1. Actual browser capture: invalid email, short password and mismatched confirmation produce field-specific errors.

The contact form is also validated. Successful submissions reset the form and show a green status message without navigating away. A successful profile screenshot is included in report/screenshots/02-validation-success.png.

Task 2 / FAQ accordion

Default hidden state, repeated toggling and smooth transitions

1. Add four service questions inside the existing Services content area.
2. Add a button for each question and a separate answer block.
3. Initialise every answer as collapsed and update its state when clicked.
4. Animate the answer height with a CSS transition.
5. Click the same question again to hide its answer.

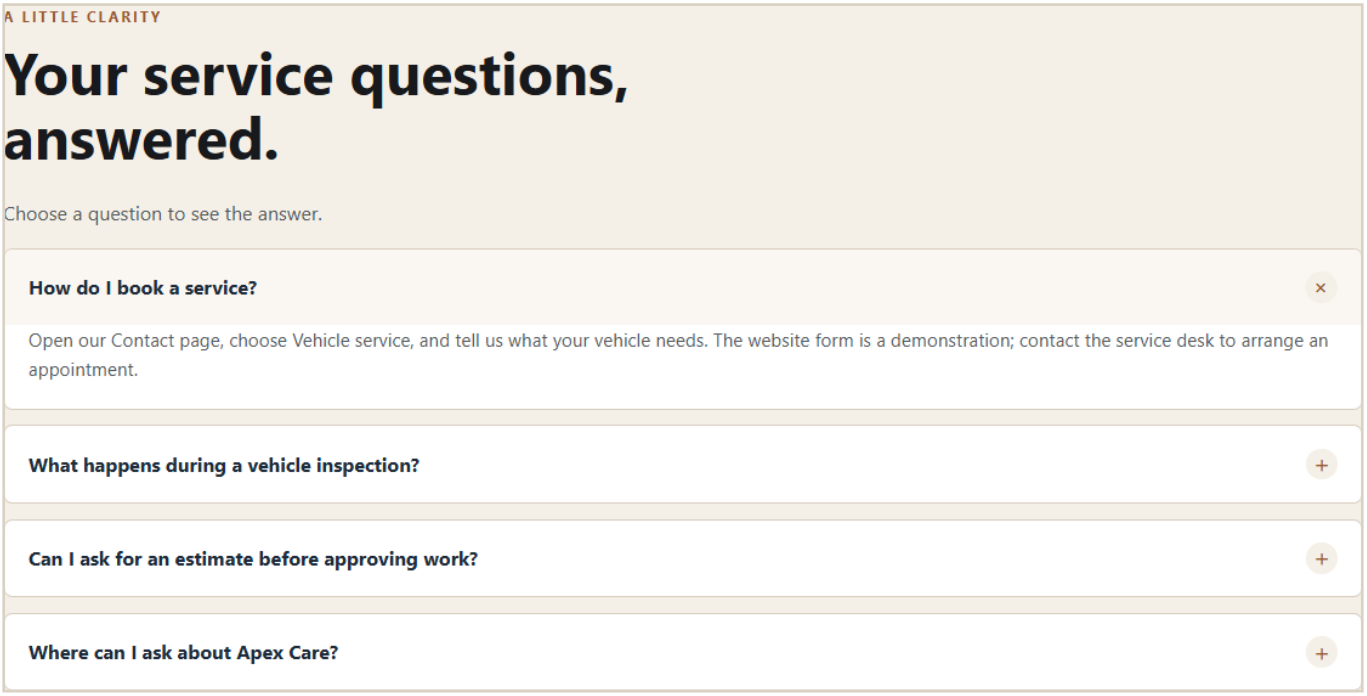


Figure 2. The first answer is expanded while the remaining three stay collapsed.

Observed result

Each question opens its own answer and closes it when clicked again. Answers are hidden by default, and the transition between the two states is smooth.

The FAQ is integrated below the original service cards. The existing service navigation, content grid, images and booking links remain in place.

Task 3 / Subscription popup

A hidden form opened by a button

1. Place a hidden subscription form and an opening button on all nine pages.
2. Use JavaScript to show the form when Subscribe to Apex is clicked.
3. Hide it when the close button is clicked.
4. Also hide it when the user clicks outside the popup.
5. Check the form on desktop and mobile screen sizes.

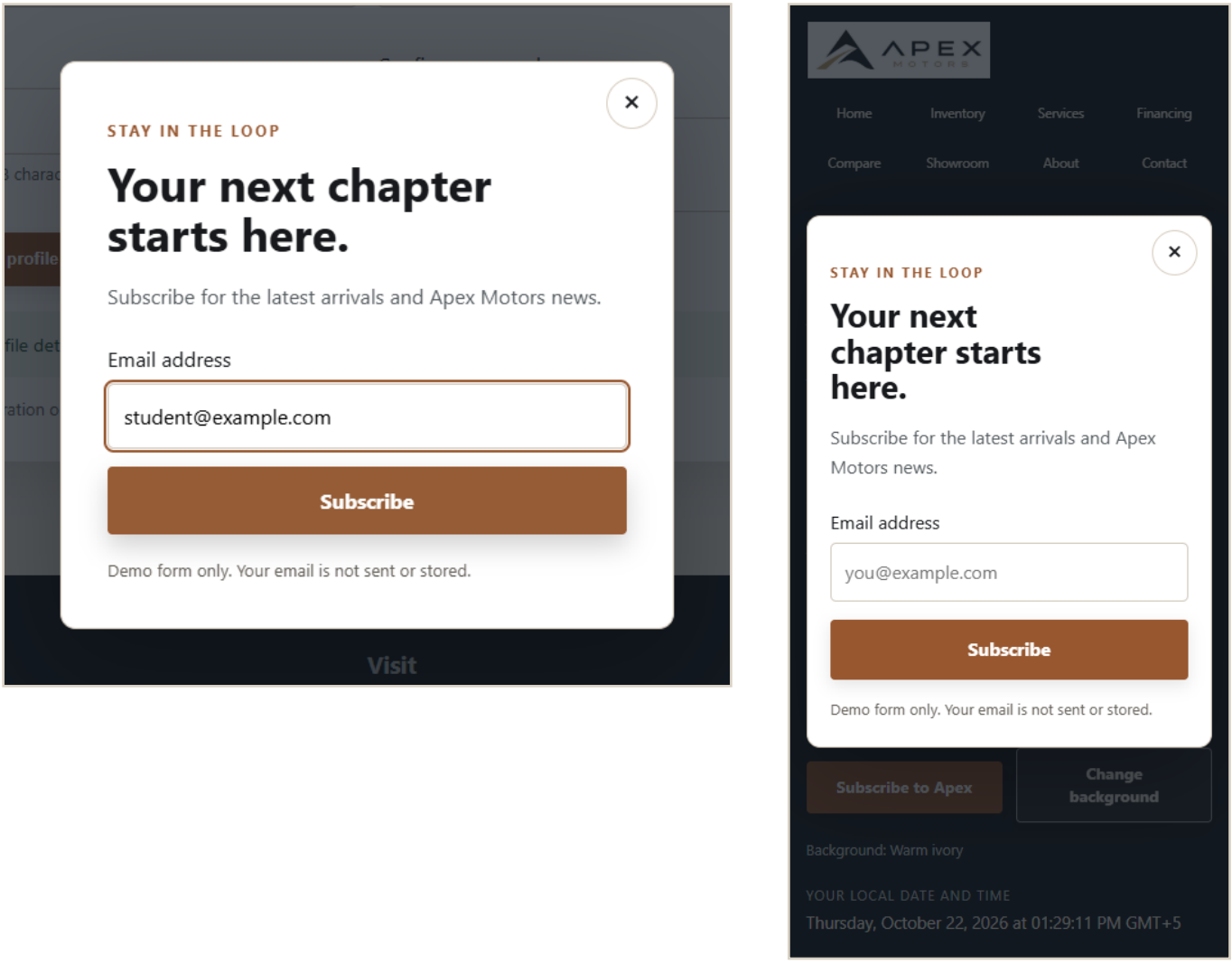


Figure 3. Actual desktop and 390 px mobile captures of the same subscription dialog.

Observed result

The popup is hidden initially. Its opening button displays the form, and both the close button and a click outside hide it again. The form fits desktop and mobile screen sizes.

Tasks 4 and 5 / Page tools

Background colors and a live local clock on every page

Task 4 steps. Add a footer button; keep a JavaScript index for three predefined colors; update document.body.style.backgroundColor on click; announce the selected color in the adjacent output. The cycle is Warm ivory, Mist blue, Soft sage, then back to Warm ivory.

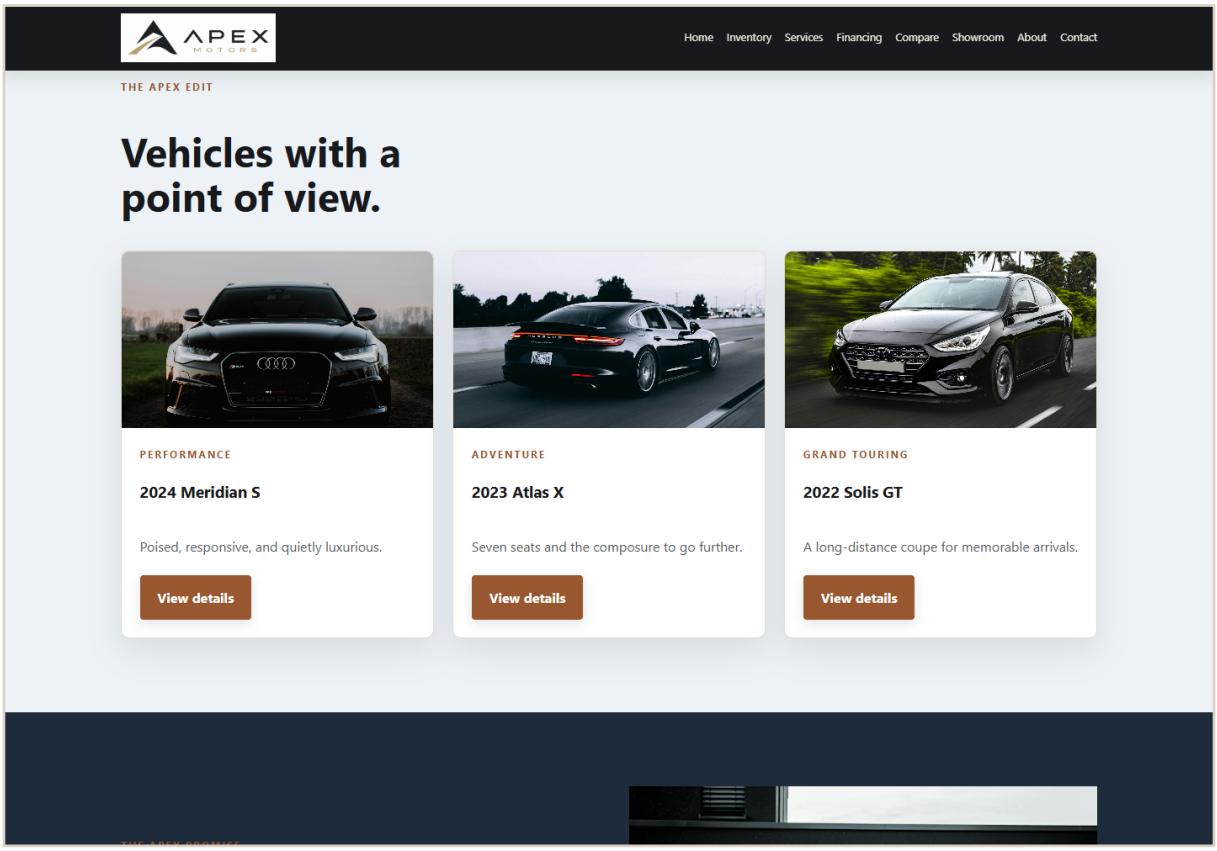


Figure 4. After pressing Change background, the existing vehicle section uses Mist blue.

Task 5 steps. Create a new Date object; format it with Intl.DateTimeFormat in readable English, including seconds and timezone; update the time element and its ISO datetime value immediately and every 1,000 milliseconds.

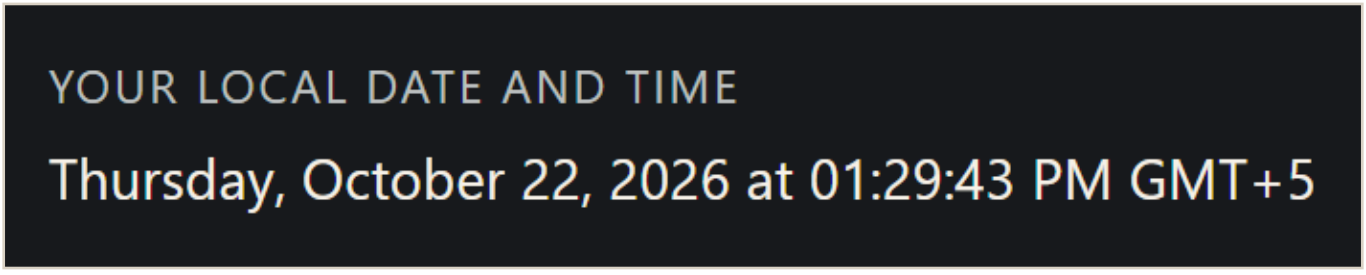


Figure 5. The live footer clock captured from the running site, including the local timezone.

The clock uses the visitor's device timezone. Test browsers used Asia/Qyzylorda. Both date and time were checked against the current browser clock, and the displayed value changed after one second.

Results and website

Task verification, team review and reflection

All five tasks were checked in Google Chrome and Microsoft Edge on desktop, tablet and mobile layouts. Required fields, email format, password length and confirmation were tested with invalid and valid input. The FAQ toggles correctly, the popup opens and closes as required, the background changes, and the date/time display updates automatically. The nine pages load their images and shared script without JavaScript errors.

Team component review

Kuttibay	Form validation: explain required checks, email pattern, minimum length and password confirmation.
Muhammad	FAQ: explain event handling, expanded state and the CSS transition.
Arsen	Popup: explain button opening, the close button and outside-click handling.
Farabi	Page tools: explain color cycling, Date, Intl.DateTimeFormat and setInterval().

Each member should understand all five components and be ready to demonstrate the website during the practical defense.

Website URL

https://vertex-blip.github.io/Car-dealership-website/

This is the existing project website. The updated project in this ZIP has been tested locally and has not been published.

Reflection

The assignment demonstrates JavaScript event handling, input validation and DOM updates in the existing Apex Motors website. Validation provides clear feedback, the accordion and popup show and hide content, the color button changes a CSS property, and Date supplies a readable current time. A shared JavaScript file keeps the same behavior across the project.